MATHS AA HL · Regression and conditional probability

Worked mark scheme · 35 minutes · 28 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

1. Can a regression be reversed? [19]

(a) **[4]**

Find the means and the regression of y on x. Use it to predict y when $x = 6.5$.

A1 · The means are 6 and 6.5.

M1 · Fit y on x by least squares.

A1 · $y = 1.21429 + 0.880952x$.

A1 · Prediction 6.94.

(b) **[4]**

Estimate x for $y = 8$ using the appropriate regression. Compare with inverting the first line.

M1 · Fit x on y.

A1 · $x = -0.413333 + 0.986667y$, giving 7.48.

M1 · Invert the first prediction line to obtain 7.70.

R1 · The least-squares residuals are measured in different coordinate directions.

(c) **[4]**

Show that the product of the two fitted slopes agrees with the square of the correlation coefficient. Explain when the two regression lines would coincide for data with non-zero variances.

M1 · Multiply 0.880952 by 0.986667.

A1 · Product is 0.8692, agreeing with 0.932312^2.

M1 · Coincidence requires reciprocal slopes, hence a product of 1.

R1 · This occurs when $|r| = 1$ and all points lie on one non-horizontal, non-vertical line.

(d) **[4]**

Let H denote an observed index at least 7 and E denote exposure at least 7. Calculate $P(H|E)$ and assess independence in this empirical distribution.

M1 · Three sites meet E and two also meet H.

A1 · $P(H|E) = 2/3$.

A1 · $P(H) = 3/6 = 1/2$.

R1 · These differ, so the events are dependent.

(e) **[3]**

A colleague substitutes the fitted values for measured y before estimating $P(H \mid E)$. Find their answer and explain the discrepancy.

M1 · At $x = 7, 8, 10$ the fitted values are approximately 7.38, 8.26 and 10.0.

A1 · The resulting fraction is 1.

R1 · Thresholding a conditional mean discards residual variation, including the observed (7,6).

2. What happens when the sample is reweighted? [9]

(a) **[3]**

Derive expressions for $P(S)$ and $P(D)$ and hence for $P(D \mid S)$.

A1 · $P(S) = 0.6w + 0.4(1 - w) = 0.4 + 0.2w$.

A1 · $P(D) = 0.24w + 0.32(1 - w) = 0.32 - 0.08w$.

M1 · Since D is contained in S, divide the second expression by the first.

(b) **[3]**

Find the choice of w for which half the selling shops offer delivery under this scheme. Compare with choosing uniformly from all 150 shops.

M1 · Equate $0.32 - 0.08w$ to $0.5(0.4 + 0.2w)$.

A1 · $w = 2/3$.

R1 · A contains two-thirds of the shops, so this matches uniform selection from all shops.

(c) **[3]**

For $0 < w < 1$ determine whether region and delivery are independent given that the shop sells the item. Justify.

M1 · Conditional delivery proportions among sellers are $24/60$ and $16/20$.

A1 · These are 0.4 and 0.8.

R1 · Both regions have positive conditional weight, so the different probabilities rule out conditional independence.

End of mark scheme · 28 marks